Curriculum Vitae

Jorge Alda Gallo
Ph.D. in Theoretical Physics

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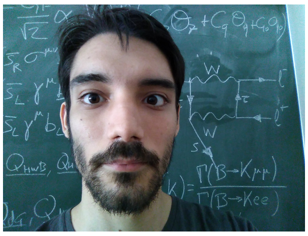

Research interests

- New physics beyond the Standard Model.
- Flavour Physics.
- B-meson anomalies.
- Effective Field Theories.
- Axions and Axion-like particles.
- Machine Learning applied to Physics.

Education

B.Sc. in Physics, Universidad de Zaragoza 2011-2015

Average grade: 9.20/10. 13 Honours.
B.Ss. Project: *"Cálculo numérico en teoría cuántica de campos de la materia condensada"*. Under the supervision of David Zueco Láinez. Qualification: 9.5/10.
Link to the B.Sc. Thesis: https://zaguan.unizar.es/record/31745

M.Sc. in Theoretical Physics, Universidad Complutense de Madrid 2015-2016

Average grade: 9.34/10.
M.Sc. Project: *New Applications of the Coleman-Weinberg Model.* Under the supervision of J. A. Ruiz Cembranos. Qualification: 9.0/10.

Ph.D. in Physics, Universidad de Zaragoza 2016-2022

Under the supervision of Siannah Peñaranda Rivas.
Dissertation title: *A Glance into Flavour Physics with Effective Field Theories and Machine Learning.*
Defense date: 27th April 2022.
Qualification: *Cum Laude.*
Link to the Ph.D. Thesis: `https://zaguan.unizar.es/record/117920`

Ph.D. School 2018

Taller de Altas Energías. Benasque (Huesca, Spain).

Ph.D. School 2022

WE-Heraeus Summer School SMEFT'22: Theory and Phenomenology of the Standard Model EFT. Siegen (Germany).

Positions

Universidad de Zaragoza April 2022 - February 2023

Università degli Studi di Padova October 2023 - September 2025

Università degli Studi di Padova December 2025 - December 2027

Contratto di Ricerca.

Grants

Grant JAE-Intro CSIC 2014

Project *"Caos semiclásico en sistemas de bosones con interacción"*, supervised by David Zueco Láinez.
CSIC-ICMA (Spanish National Research Council and Instituto de Ciencia de Materiales de Aragón).

PreDoc Grant, Diputación General de Aragón 2017-2022

Programa Ibercaja-CAI de Estancias de Investigación 2021

Grant No. CB 5/21.

Postdoctoral Scholarship for Extension of Studies Abroad for Life and Material Sciences, Ramón Areces Foundation 2023-2025

Memberships

CAPA 2019-present

Centro de Astropartículas y Física de Altas Energías. Zaragoza, Spain.
`capa.unizar.es`

INFN 2023-present

Associated to Istituto Nazionale di Fisica Nucleare, Sezione di Padova.

COMCHA 2026-2028

Spanish Network on Advanced Computing for Fundamental and Applied Physics.
Convener of WG2: Simulation and Modelling.

Visiting Scholar Appointments

Università degli Studi di Padova/INFN Summer 2021

Università degli Studi di Padova/INFN Summer 2023

University of California, San Diego September 2024

Laboratori Nazionali del Gran Sasso November 2024

Kavli IPMU, Tokyo May-June 2026

Scientific Production

J. Alda, J. Guasch and S. Peñaranda: ***Some results on Lepton Flavour Universality Violation***
Eur.Phys. J. C, 79 7 (2019) 588

doi:10.1140/epjc/s10052-019-7092-x
arXiv:1805.03636 [hep-ph]

J. Alda, J. Guasch and S. Peñaranda: ***Anomalies in B decays: A phenomenological approach***
Eur.Phys. J. Plus 137 (2022) 217
doi:10.1140/epjp/s13360-022-02405-3
arXiv:2012.14799 [hep-ph]

J. Alda, J. Guasch and S. Peñaranda: ***Anomalies in B decays: Present status and future collider prospects***
arXiv:2105.05095 [hep-ph]
SLAC eConf C21-03-15.1

J. Alda, J. Guasch and S. Peñaranda: ***Using Machine Learning techniques in phenomenological studies in flavour physics***
J. High Energ. Phys. 2022, 115 (2022)
doi:10.1007/JHEP07(2022)115
arXiv:2109.07405 [he-ph]

J. Alda, J. Guasch and S. Peñaranda: ***Exploring B-physics anomalies at colliders***
arXiv:2110.12240 [hep-ph]
PoS(EPS-HEP2021)494

J. Alda, A. W. M Guerrera, S. Peñaranda and S. Rigolin: ***Leptonic Meson Decays into Invisible ALP***
Nucl.Phys.B 979 (2022) 115791
doi:10.1016/j.nuclphysb.2022.115791
arXiv:2111.02536 [hep-ph]

J. Alda, G. Levati, P. Paradisi, S. Rigolin and N. Selimović: ***Collider and astrophysical signatures of light scalars with enhanced τ couplings***
JHEP 06 (2025) 008
doi:10.1007/JHEP06(2025)008
arXiv:2407.18296 [hep-ph]

S. Peñaranda, J. Alda and A. Mir: ***Flavour Anomalies: A comparative Analysis using a Machine Learning Algorithm***
Int.J.Theor.Phys. 65 (2026) 2, 46
doi:10.1007/s10773-026-06263-y
arXiv:2412.15830 [hep-ph]

J. Alda, C. Broggini, G. Di Carlo, L. Di Luzio, D. Piatti, S. Rigolin and C. Toni: ***Weak nuclear decays deep-underground as a probe of axion dark matter***
Phys.Rev.D 111 (2025) 3, 035022
doi:10.1103/PhysRevD.111.035022

arXiv:2412.20932 [hep-ph]

J. Alda, M. Fuentes Zamoro, L. Merlo, X. Ponce Díaz and S. Rigolin: ***Comprehensive ALP Searches in Meson Decays***
arXiv:2507.19578 [hep-ph]
Submitted to Fortschritte der Physik

J. Alda, M. Fuentes Zamoro, L. Merlo, X. Ponce Díaz and S. Rigolin: ***ALPaca: The ALP Automatic Computing Algorithm***
arXiv:2508.08354 [hep-ph]
Submitted to Computer Physics Communications

A. Mir, J. Alda and S. Peñaranda: ***B-Meson Anomalies: Effective Field Theory Meets Machine Learning***
arXiv:2510.17742 [hep-ph]
Contribution to EPS-HEP2025

M. Fuentes Zamoro, J. Alda, L. Merlo, X. Ponce Díaz and S. Rigolin: ***Tackling ALP searches in meson decays with ALPaca: a phenomenological approach***
Proceedings of 3rd General Meeting of the COST Action COSMIC WISPers (CA21106)
`https://pos.sissa.it/507/057`

J. Alda: ***Lecture notes on Machine Learning applications for global fits***
arXiv:2604.07520 [hep-ph]
Submitted to SciPotst Physics Lectures Notes

J. Alda, J. Asorey, A. Mir, S. Peñaranda: ***Physically Consistent Parameter Inference: Transparent Machine Learning Emulation in High Energy Physics and Cosmology***
Universe. 2026; 12(8):238
doi:doi.org/10.3390/universe12080238
arXiv:2607.12726 [hep-ph]
Invited contribution to Special Issue "Advances in Machine Learning Techniques in Cosmology and Particle Physics".

J. Alda, L. C. Bresciani, G. Levati, P. Paradisi, S. Rigolin and N. Selimović: ***Leptonic Couplings of Axion-Like Particles: Constraints from Present and Next-Generation Colliders***
In preparation

J. Alda, S. Rigolin: ***ALPs in Chiral Perturbation Theory: general framework and baryon interactions***
In preparation

Talks and conferences

2nd Red LHC Workshop. Madrid, Spain. 9-11 May 2018
Talk "Some Results on Lepton Flavour Violation".

Taller de Altas Energías. Benasque (Huesca, Spain) 2-15 September 2018
Talk “Some Results on Lepton Flavour Violation”.

X CPAN Days. Salamanca, Spain. 29-31 October 2018
Talk “Complex Wilson coefficients in the analysis of B-anomalies”.

I Jornadas de Jóvenes Investigadores CAPA. Zaragoza, Spain. 7 May 2019
Invited Talk “Effective Theories for B-meson anomalies”.

I Jornadas del Programa de Doctorado de Física. Zaragoza, Spain. 20 June 2019
Invited Talk “Effective Theories for B-meson anomalies”.

XXXVII Bienal de Física de la Real Sociedad Española de Física. Zaragoza, Spain. 15-19 de July 2019
Talk “Some Results on Lepton Flavour Universality Violation”.

International Workshop on Future Linear Colliders - LCWS2021. Online. 15-18 March 2021
Talk “Anomalies in B mesons decays: Present status and future collider prospects”.

European Physical Society Conference on High Energy Physics 2021 (EPS-HEP2021). Online. 26-30 July 2021
Poster “Exploring B-physics anomalies at colliders ”.

Seminar, Department of Theoretical Physics. University of Zaragoza, Spain. 18 November 2021
Invited Talk “Leptonic Mesons Decays into invisible ALP”.

II Jornadas del Programa de Doctorado de Física. University of Zaragoza, Spain. 3 December 2021
Invited Talk “Leptonic Mesons Decays into invisible ALP”.

Seminar, Department of Theoretical Physics. University of Zaragoza, Spain. 20 January 2022
Invited Talk “Using Machine Learning techniques in phenomenological studies in flavour physics”.

Seminar, Institute of Theoretical Physics (IFT). Autonomous University of Madrid, Spain. 27 January 2022
Invited Talk “Using Machine Learning techniques in phenomenological studies in flavour physics”.

XIII CPAN Days. Huelva, Spain. 21-23 March 2022
Talk “Using Machine Learning techniques in phenomenological studies in flavour physics”.

5th Inter-experiment Machine Learning Workshop. CERN, Switzerland. 8-13 May 2022
Talk “Using Machine Learning techniques in phenomenological studies in flavour physics”.

Saturnalia Workshop. University de Zaragoza/CAPA, Spain. 21 December 2022
Invited Talk “Using Machine Learning techniques in phenomenological studies in flavour physics”.

Saturnalia Workshop. University de Zaragoza/CAPA, Spain. 20 December 2023
Invited Talk “Light New Physics coupling to τ”.

Seminar, Institute of Theoretical Physics (IFT). Autonomous University of Madrid, Spain. 06 June 2024
Invited Talk “Leptophilic Axion-Like Particles”.

Seminar, University of California, San Diego. 17 September 2024
Invited Talk “Signatures of light scalars with enhanced τ couplings”.

Flavour for New Physics at Present and Future Colliders Workshop. Mainz Institute for Theoretical Physics, Johannes Gutenberg University, Germany. 19 June 2025
Invited Talk “Present and future of ALP searches in colliders”.

Invisibles25 Workshop. CERN, Switzerland. 4 September 2025
Talk “ALP searches in meson decays with ALPaca: a phenomenological approach”.

Implications of LHCb measurements and future prospects. CERN, Switzerland. 5 November 2025
Invited Talk “ALP searches in meson decays with ALPaca”.

Saturnalia Workshop. University de Zaragoza/CAPA, Spain. 18 December 2025
Invited Talk “GeV-ALP searches with ALP-aca”.

4th COMCHA School. University of Zaragoza/CAPA, Spain. 14 April 2026
Invited Talk and hands-on session “ML for global fits”.

Invisibles26 Workshop. A Coruña, Spain. 10-14 August 2026
Poster and Talk “GeV ALP interactions with hadrons”.

Contributions to public repositories

flavio

1 pull request merged: `https://github.com/flav-io/flavio/pull/160`

smelli

1 pull request: `https://github.com/smelli/smelli/pull/45`

ALP Automatic Computing Algorithm (ALP-aca)

Main contributor. `https://github.com/alp-aca`

Open source Python library for the phenomenology of Axion-Like Particles (ALPs) in meson decays. It provides a framework to match from specific UV models, run the RGE equations, compute the branching ratios of meson decays into ALPs and the corresponding experimental constraints, and set exclusion regions.

Teaching

September 2019

Ph.D. School "Taller de Altas Energías de Benasque" (Huesca, Spain).
Associate teacher.

2019-2020

Differential Equations
Problem-solving sessions, 38 teaching hours.
Second year course, Bachelor Degree in Physics, Universidad de Zaragoza.

General Physics
Laboratory sessions, 10 teaching hours.
First year course, Bachelor Degree in Mathematics, Universidad de Zaragoza.

2020-2021

Differential Equations
Problem-solving sessions, 38 teaching hours.
Second year course, Bachelor Degree in Physics, Universidad de Zaragoza.

General Physics
Laboratory sessions, 10 teaching hours.
First year course, Bachelor Degree in Mathematics, Universidad de Zaragoza.

2021-2022

Differential Equations
Problem-solving sessions, 38 teaching hours.
Second year course, Bachelor Degree in Physics, Universidad de Zaragoza.

Co-direction of B.Sc. Project: Alejandro Mir
10 teaching hours. Fourth year course, Bachelor Degree in Physics, Universidad de Zaragoza.
Link to the B.Sc. Thesis: `https://zaguan.unizar.es/record/125482`

General Physics
Laboratory sessions, 10 teaching hours.
First year course, Bachelor Degree in Mathematics, Universidad de Zaragoza.

2022-23

Co-direction of M.Sc. Project: Alejandro Mir
10 teaching hours. Master Degree in Physics, Universidad de Zaragoza.
Link to the M.Sc. Thesis: `https://zaguan.unizar.es/record/133532`

Co-direction of B.Sc. Project: Francisco J. Gómez-Fauro
10 teaching hours. Fourth year course, Bachelor Degree in Physics, Universidad de Zaragoza.
Link to the B.Sc. Thesis: `https://zaguan.unizar.es/record/127238`

2024-present

Co-direction of Ph.D. Thesis: Alejandro Mir
Co-direction of Ph.D. Thesis in Physics, Universidad de Zaragoza.

September 2026

Co-direction of Master Thesis: Filippo Cucchetto
Co-direction of Master Thesis in Physics, Università degli Studi di Padova.

Scientific Dissemination

Dark Matter Day. University of Zaragoza, Spain. 31 October 2022

Organizing Committee of the 2022 Saturnalia Workshop. University of Zaragoza/CAPA, Spain. 16-22 December 2022
`https://indico.capa.unizar.es/event/32/`

Local Organizing Comitte of PLANCK2025 - The 27th International Conference From the Planck Scale to the Electroweak Scale. Padova, Italy. 26-30 May 2025
`https://indico.dfa.unipd.it/event/1200/`

4th Training School on Computing Challenges, COMCHA network. University of Zaragoza/CAPA, Spain. 8-15 April 2026
`https://indico.capa.unizar.es/event/42/`
Topical lecture and hands-on session “ML for global fits”.

Junior Organizing Comittee of Invisibles26. A Coruña, Spain, 10-14 August 2026
`https://indico.cern.ch/event/1561367/`

Languages

Spanish	■■■■■	
English	■■■■■	
Italian	■■■■□	[C1 certification]
German	■■■□□	
French	■■□□□	

Programming languages

TeX/LaTeX	■■■■■
Python	■■■■■
C/C++	■■■□□
Mathematica	■■■□□
Docker	■■□□□
Rust	■□□□□

Awards

XXII Spanish Physics Olympiad 2011

Silver Medal (Rank 15).
Second position in the regional Aragonese phase.

20th International Mathematics Competition 2013

Bronze Medal (rank 177).

About this CV

This CV was last updated in September 2, 2026.
You can find the latest version at
`https://github.com/Jorge-Alda/Jorge-Alda/releases/latest/download/CV.pdf`